

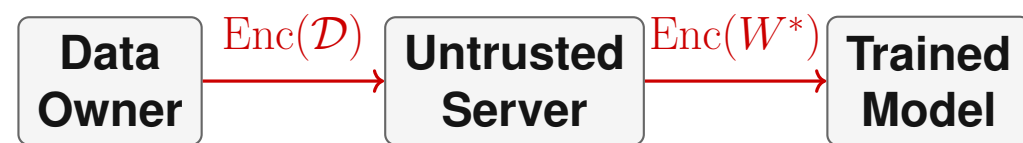
Evaluation of Loss Functions under Fully Homomorphic Encryption for Multiclass Classification

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Problem Statement

Private Training as a Service: a data owner encrypts dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ and delegates model training to an untrusted server. The server operates exclusively on ciphertexts.



Key design choice: the *loss function* dictates which non-linear operations the server must evaluate homomorphically. No prior work has systematically studied this trade-off.

What are the FHE efficiency vs. accuracy trade-offs between square loss and cross-entropy for multiclass private training?

Background: CKKS Fully Homomorphic Encryption

The **CKKS** scheme [1] supports approximate arithmetic over encrypted vectors (up to $N_0=2^{14}$ or 2^{15} values packed into one ciphertext via *SIMD slots*).

- Multiplicative depth (levels):** each homomorphic multiplication (MulRelin + Rescale) consumes one level. The level budget is fixed by the parameter set.
- Bootstrapping** refreshes spent levels at the cost of additional evaluation keys (~ 60 s per ciphertext).
- SIMD parallelism:** LogN=15 packs 16,384 values per ciphertext; all slots undergo the same arithmetic simultaneously.
- Non-linear functions** (exp, $x^{-1/2}$, comparison) require polynomial approximations; every polynomial evaluation consumes multiple levels.

CKKS multiplicative level budget (example: 21 levels)

20 18 16 14 12 10 8 6 ... operations use levels

Related Work

- Hui & Belkin (ICLR'21)** [2]: square loss matches cross-entropy in the *plaintext* setting for classification, motivating the FHE comparison.
- Chen et al. (2018)** [3]: encrypted logistic regression under FHE (*binary* only, no multiclass softmax).
- Graepel et al. (2012)** [4]: linear classifier under FHE using squared loss; no softmax, multiclass via one-vs-all only.
- Cho et al. (CCS'24)** [5]: fast FHE-friendly softmax via Normalize-and-Square, which enables the CE training evaluated here.

Gap: no prior work compares SQ vs. CE for *multiclass* private training under a unified FHE framework.

Square Loss under FHE

Loss: $\mathcal{L}_{SQ}(\mathbf{z}, y) = \|\mathbf{e}_y - W\mathbf{x}\|_2^2$

Gradient for weight w_{kj} : $\nabla_{w_{kj}} \mathcal{L}_{SQ} = \frac{2}{N} \sum_{i=1}^N (z_k(\mathbf{x}_i) - (e_y)_k) x_{ij}$

FHE-Friendly Properties

- No polynomial approximation:** gradient is a pure inner product (only multiplications and additions).
- Shallow circuit:** 4 Rescale operations per epoch $\Rightarrow$ only 5 CKKS levels needed.
- Real bootstrapping** used for level refresh: Lattigo v6 CKKS bootstrapper, K calls per epoch.

Operation (per epoch)	Level in	Level out
Forward MulRelin + Rescale	L_5	L_4
Mask slot-0 MulNew + Rescale	L_4	L_3
Gradient MulRelin + Rescale	L_3	L_2
Scale step MulNew + Rescale	L_2	L_1
<i>Bootstrap (level refresh)</i>	L_1	L_5

Total: 4 levels consumed. 5 levels suffice.

Cross-Entropy via Normalize-and-Square Softmax

Loss: $\mathcal{L}_{CE}(\mathbf{z}, y) = -\log\left(\frac{e^{z_y}}{\sum_{k=1}^K e^{z_k}}\right) = -\log \hat{p}_y$

Gradient: $\nabla_{w_{kj}} \mathcal{L}_{CE} = \frac{1}{N} \sum_i (\hat{p}_k(\mathbf{x}_i) - (e_y)_k) x_{ij}$

Challenge: computing softmax($\mathbf{z}$) in FHE requires approximating exp and $(\cdot)^{-1}$, each over a poorly-known interval, consuming many levels.

Normalize-and-Square Algorithm (Cho et al., CCS 2024)

Input: $\mathbf{z} \in [-M, 0]^K$ (logit-shifted s.t. $\max_k z_k = 0$), $k=1$ iteration

Output: $\hat{\mathbf{p}} \approx \text{softmax}(\mathbf{z})$

- $y_k \leftarrow \widetilde{\exp}(z_k/4)$ *deg-7 poly, $L_{21} \rightarrow L_{17}$*
- $S \leftarrow \sum_{k=1}^K y_k^2$ *$K-1$ additions, no level cost*
- $\lambda \leftarrow S^{-1/2}$ *deg-5 poly, $L_{17} \rightarrow L_{13}$*
- $\hat{p}_k \leftarrow (\lambda \cdot y_k)^2$ *normalize & square, $L_{13} \rightarrow L_{11}$*

Total depth: 10 levels per softmax call.

SIMD Batch Packing

All N training samples are packed into CKKS slots: $X_{cl}^{(j)}[\text{slot } i] = x_{ij}$. This reduces polynomial evaluations from $N \times K$ to K per epoch, an N -fold saving.

Experimental Setup

- Scheme:** CKKS via Lattigo v6 (Go). SQ: LogN=14, LogQ=[55, 45×5], 5 usable levels. CE: LogN=15, LogQ=[55, 40×21], 21 usable levels.
- Bootstrapping:** SQ uses real CKKS bootstrapper; CE uses emulated bootstrap (decrypt + re-encrypt), targeting future deployment with real bootstrap on cloud hardware.
- Datasets:** three UCI benchmarks (standardized features, stratified 80/20 split).
- Training:** full-batch gradient descent with encrypted weights, SIMD-packed ciphertext columns.

Results

FHE Circuit Complexity

Metric	Square Loss	Cross-Entropy
Max. multiplicative depth	5	21
Levels consumed / epoch	4	≥ 10 (softmax)
Polynomial approximations	None	Exp (deg 7) + $x^{-1/2}$ (deg 5)
Bootstrap calls / epoch	K	$K + K \times F$
Time / epoch, Iris ($N=120$)	~ 35 s	~ 668 s [†]
Time / epoch, Wine ($N=144$)	~ 145 s	est. ~ 800 s
Time / epoch, Glass ($N=174$)	~ 242 s	est. $> 1,900$ s

[†]Sample-by-sample CE, Iris epoch 0 measured. CE cost scales as $\mathcal{O}(N \cdot K)$ polynomial evaluations per epoch.

Datasets & Test Accuracy

Dataset	N	F	K	CE Acc.	SQ Acc.
Iris	150	4	3	86.7%	82.5%
Wine	178	13	3	94.1%	94.1%
Glass	214	9	6	42.5%*	47.5%

*CE Glass used sub-optimal LR; hyperparameter re-tuning in progress.

Key Findings

- SQ is more FHE-efficient:** 5 levels vs. 21 for CE.
- CE matches SQ accuracy** on (Wine 94.1%), at $\sim 19\times$ higher training cost per iter.
- Softmax is the bottleneck:** Normalize-and-Square consumes 10 of CE's 21 levels.

References

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